

Anuj Patil

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Summary

Final-year IT Backend Engineer with AI Systems Experience. Designed CodeEcoScan, a live static analysis tool that quantifies the carbon footprint of Python code; built Agent Sentinel, an autonomous AI agent for real-time misinformation detection (presented at Mumbai Hacks 2025); and architected RAG-based document systems with vectorised PostgreSQL, Redis caching, and Docker microservices. Comfortable owning full-stack projects end-to-end — Spring Boot and FastAPI backends to React frontends — with hands-on experience in LLM fine-tuning (LoRA), LangChain pipelines, and production deployments on Railway and Render.

Experience

Software Development Intern (Remote) 2026 – Present

[Codologs] — Navi Mumbai / Remote

- Backend Engineering: Built scalable microservices using Spring Boot 3 and designing and optimizing RESTful APIs.
- Implemented authentication + session management service.
- Reduced API latency by using Redis caching
- Built 3 microservices used by production system

Education

Bachelor of Technology — Computer Engineering 2022 – 2026

Pillai College of Engineering, University of Mumbai · CGPA: 7.7 / 10.0

Technical Skills

Languages: Java, Python, JavaScript, TypeScript, C

Backend: Spring Boot, FastAPI, Node.js, REST API Design

Frontend: React.js, HTML/CSS

Databases: PostgreSQL (pgvector), MySQL, Redis

AI / ML: LangChain, RAG, LoRA Fine-Tuning, ChromaDB

DevOps / Cloud: Docker, Docker Compose, GitHub Actions, AWS (S3, EC2 basics), Railway, Render

Tools: Git, IntelliJ IDEA, VS Code, Jupyter Notebook, Postman, Sentry

Core CS: Data Structures & Algorithms, OOP, System Design fundamentals, Multithreading

Projects

CodeEcoScan — Lead Developer · [Live public API](#)

- Engineered a static analysis tool that estimates energy risk and carbon footprint of Python workloads, scoring code across loop complexity, recursion depth, import weight, and I/O patterns using a custom AST-based engine.
- Architected a real-time carbon emissions estimator parameterised by hardware profile (Laptop / Cloud VM / GPU Server), runtime, and run frequency — converting energy draw to kgCO₂/day.
- Deployed a live public API on Render; built a terminal-inspired React frontend with syntax-highlighted heatmap overlays and per-category score breakdowns.

Tech: Python (FastAPI), React.js, AST analysis, Render

Agent Sentinel — Lead Backend Engineer · Presented at Mumbai Hacks 2025

- Designed and built a fully autonomous AI agent that detects, verifies, and neutralises viral misinformation in near real-time targeting Government (B2G) and Enterprise (B2B) use cases.
- Implemented a multi-stage pipeline: claim extraction → GDELT-backed fact retrieval → RoBERTa-based classification → structured verdict generation with confidence scores.
- Selected to present at Mumbai Hacks 2025 (December) — one of Maharashtra's largest hackathons — competing among 200+ teams as a 2-member team.

Tech: Python, FastAPI, HuggingFace Transformers (RoBERTa), GDELT API, Docker

SmartDoc Query Engine — Full-Stack Developer / DevOps

- Architected a production-grade RAG system for secure PDF querying using a microservice architecture with Docker Compose, achieving high availability through containerised service isolation.
- Implemented vectorised document storage with PostgreSQL (pgvector) and Redis-based caching + rate limiting, reducing repeat query latency by eliminating redundant embedding calls.
- Added full observability via Sentry with structured logging of model performance and edge-case parsing failures.

Tech: Python (FastAPI), React.js (Vite), PostgreSQL (pgvector), Redis, Docker Compose, Sentence Transformers

AI Tutor with Voice Cloning & Persona Preservation — Backend Developer (4-member team)

- Built a custom-trained AI tutor that answers course-specific questions while preserving the instructor's unique persona, achieving 95% response accuracy on course-specific evaluation benchmarks.
- Integrated voice cloning (XTTS) with RAG-based retrieval over ChromaDB, enabling the tutor to respond in the instructor's synthesised voice using LoRA fine-tuned models.

Tech: Python, FastAPI, Streamlit, RAG, LoRA Fine-Tuning, XTTS (Voice Cloning), ChromaDB, MLX

Changelog-AI [Live Public API](#)

- Developed an end-to-end NLP pipeline using Gemini models and LangChain to transform raw GitHub commit metadata into structured, user-centric release notes automatically.
- Deployed a high-availability backend on Railway with Sentry observability — monitoring model performance and capturing edge-case parsing failures in real-time.

Tech: Python, LangChain, Gemini API, Railway, Sentry

Password Manager CLI

- Built a secure command-line password manager using a Doubly Linked List as the in-memory data structure, AES encryption for credential storage, and clipboard expiry to prevent clipboard sniffing attacks.

Tech: Java, AES Encryption, Security

Hackathons & Achievements

Mumbai Hacks 2025 — Participant December 2025

Mumbai, Maharashtra

- Built and presented Agent Sentinel, an autonomous misinformation detection system, as a 2-member team at one of Maharashtra's largest annual hackathons.

Smart India Hackathon (SIH) — Internal Round Participant 2025

Pillai College of Engineering, New Panvel

- Cleared the internal college selection round.